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Chapter 1:
Introduction

The atmosphere is a chaotic dynamic system with many phenomena happening. For a long time, researchers have been trying to model the atmosphere for weather forecasting and climate modelling. Among them, the gradient of wind velocity is important. In the past, the wind velocity was measured using radiosonde, a balloon-borne instrument released once now and then, at a fixed location. Although radiosondes remain the reference standard, they provide sparse temporal sampling, drift away from the launch site as they ascend, and cannot offer a continuous view of evolving atmosphere. In this context, remote sensing by wind profile radar provides a powerful complement.

1.1 Wind profile radar

A wind profiler radar transmits electromagnetic pulses into the sky and picks up extremely weak echoes returned by minute fluctuations in the refractive index of air produced by turbulence and hydrometeors such as raindrops. Because these scatterers are carried by the wind, the Doppler shift of the returned signal can be used to estimate the velocity of the air along the radar beam. By pointing the beam in different directions, the 3D wind vector can be reconstructed at every range gate, giving a near-continuous record of the wind. The beams can be steered mechanically or electronically. This process is called Doppler Beam Swing (DBS).

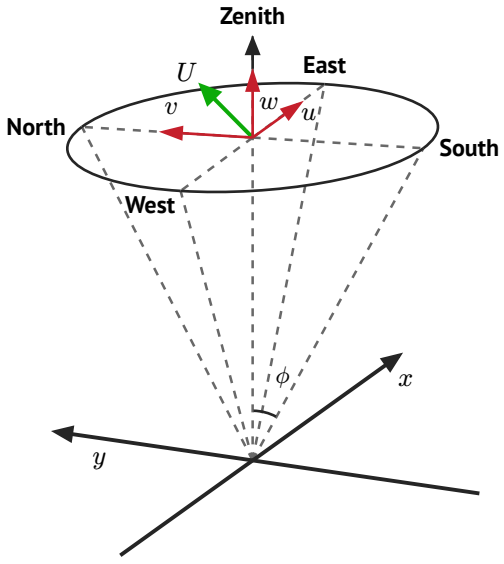


Figure 1.1: Doppler Beam Swing method

However, the quality of the record depends entirely on how reliably the wind velocity is estimated from the measured Doppler spectrum. In this report, we discuss few algorithms used in velocity estimation, as well as a test bench upon which algorithms can be further evaluated and developed. The main focus is on processing radar data obtained from Stratosphere–Troposphere radar (ST Radar) the operating at the Cochin University of Science and Technology, Kochi.

1.2 The ST Radar Facility at CUSAT

The ST radar is located at the Advanced Centre for Atmospheric Radar Research (ACARR) on the university campus in Kochi. It was proposed to the Department of Science and Technology, Government of India, under its IRHPA scheme, with the specific aim of studying the characteristics and dynamics of the Indian summer monsoon at its gateway. Kerala is the point at which the south-west monsoon first makes landfall over the Indian subcontinent, and a continuously operating wind profiler at this location provides a unique vantage point on the onset, progress and intraseasonal variability of the monsoon circulation, as well as on other weather phenomena. The radar was designed, developed and installed indigenously, with the system fabricated by Data Patterns under the guidance of the university's scientists and engineers, and it became fully operational in December 2016. It is regarded as the first atmospheric wind profiler radar to operate successfully in the 205 MHz frequency band.

1.2.1 Basic specifications

The specifications of the radar are summarised in Table 1.1

Parameter	Specification
Frequency	205 MHz
Bandwidth	5 MHz
Type of system	Active Phased Array
Antenna element	Three Element Yagi-Uda
Height coverage	315 m to 20 km
Range gates	1024 (Programmable)
Modes	DBS/SAM
Height resolution	$\approx$ 45 m to 720m
Beam width	$\approx 3°$
Off Zenith Angle	$0°$ to $30°$ in steps of $1°$
Beam positions in Azimuth	$0°$ - $360°$ with $1°$ resolution
Pulse width	$0.3\mu s$ to $76.8\mu s$
Modulation	Binary Phase Shift Keying(BPSK)
Code	Complementary Code/Barker Code
Baud rate	$0.3\mu s$ to $4.8\mu s$ in steps of 0.3
PRF	100 Hz to 16 kHz Selectable
Peak Power	500 W (typical per element)
Duty ratio	Up to 15 % (Max)
Peak power aperture product	$\approx 1.6 \times 10^8$ (typical) Wm^2
Radar system sensitivity	-165 dBm(post processing)
Dynamic range	70 dB(min)
Type of receiver	Direct band with sampling
Type of signal processor	FFT based frequency domain

Parameter	Specification
Horizontal wind Velocity	Up to 70 ms^{-1}
Vertical wind velocity	Up to 30 ms^{-1}
Radial velocity resolution	Better than 0.1 ms^{-1}

Table 1.1: Specification of ST Radar from [1]

The radar has been validated using measurements from radiosonde. The caliberation is performed using radio source Virgo A and by tracking the Moon

1.3 Signal processing chain

The radar signal processor converts the raw received voltages into geophysical products in a sequence of well-established steps. Complex (in-phase and quadrature) samples are collected for each range gate and beam direction; successive pulses are coherently integrated to improve the signal-to-noise ratio and reduce the data rate; a time series is formed at each gate; and a spectral transform, usually a Fast Fourier Transform, yields the Doppler power spectrum. Ground clutter and other interference are suppressed, the noise floor is estimated, and from the remaining signal spectrum three quantities are derived. These are the zeroth moment, proportional to the received signal power; the first moment, which gives the mean Doppler shift and hence the radial velocity; and the second moment, which gives the spectral width and is related to turbulence intensity. The radial velocities from the different beam directions are finally combined to produce the zonal, meridional and vertical components of the wind. Every one of these final products rests upon the correctness of the moment estimates.

1.4 Existing Challenges in Wind Profiler Data Processing

1.4.1 Weak signals and low signal-to-noise ratio

Atmospheric backscatter at VHF is extraordinarily weak, and its strength decreases rapidly with height as the air becomes drier and the turbulence less intense. In the upper troposphere and lower stratosphere the echo can sink to, or below, the level of the receiver and sky noise. In the Doppler spectrum, the atmospheric peak then becomes only marginally higher than the fluctuations of the noise floor, and the estimation of its position and width becomes uncertain.

1.4.2 Contamination of the spectrum

The Doppler spectrum is rarely a clean, single Gaussian peak. Several forms of contamination commonly appear, particularly in a tropical coastal location such as Kochi:

- Ground clutter and fixed echoes from nearby buildings, trees and terrain, concentrated near zero Doppler and capable of masking genuine low-velocity signals.
- Radio-frequency interference and spurious spikes from other transmitters and from electronic equipment, which show up as narrow, strong lines at arbitrary frequencies.
- Aircraft, birds and insects, which produce transient, strong echoes that bear no relation to the wind and which can easily be mistaken for the true atmospheric peak.

- Precipitation, which in the monsoon can add a second peak to the spectrum, since raindrops fall relative to the air; the radar then sees both the clear-air signal and the hydrometeor signal at once.
- Multiple atmospheric layers within a single range gate, producing multi-peaked or broadened spectra.

1.4.3 Ambiguity in peak identification

When the spectrum contains more than one local maximum, the strongest peak is not always the correct one, and a simple search for the highest value can lock onto interference. Conversely, if the true signal is weak, a too-conservative threshold will discard it. The result of a wrong choice is an outlier in the velocity profile: a sudden, unphysical jump in wind speed at a single height, which then propagates into derived quantities such as shear, momentum flux and kinetic energy.

1.4.4 Computational and operational demands

A phased-array wind profiler running continuously generates very large volumes of data. Methods applied to this data need to be fast enough to operate in near real time, robust enough to run without constant human supervision, and consistent across seasons and atmospheric conditions. Manual inspection of spectra, which is feasible for a short campaign, is not a viable option for a multi-year archive.

1.5 Existing Solutions and Their Limitations

The wind profiler community has developed a range of approaches to these difficulties over several decades. They can be grouped broadly into the following families.

Direct moment computation with thresholding. The simplest approach computes the moments of the spectrum directly over a window around the strongest peak, after subtracting the estimated noise level. It is fast and straightforward, but it depends heavily on a good noise estimate and a sensible choice of integration limits, and it fails when interference or a second peak lies inside the window.

Gaussian fitting. Here a Gaussian curve is fitted to the signal peak, from which the power, mean Doppler and width are obtained as the fitted parameters. This approach resists noise better than raw moments and gives physically meaningful widths, but it assumes a single Gaussian-shaped peak, it is iterative and therefore more costly, and its result is sensitive to the starting guess. For multi-peaked spectra it may converge to the wrong feature.

Spectral averaging and filtering. Incoherent averaging of consecutive spectra reduces noise variance and makes the true peak stand out, and median or similar filters can remove isolated outliers from the resulting velocity profiles. The cost is reduced time resolution, and the fact that post-hoc filtering cannot recover information that has already been misidentified.

Continuity and consensus methods. Procedures that exploit the continuity of the wind field in height and time, or that apply consensus averaging across neighbouring estimates, can reject outliers effectively. They do, however, rely on the assumption that most estimates in a neighbourhood are correct, and in conditions of rapidly changing wind or extensive contamination that assumption may fail.

Advanced and adaptive techniques. More recent work employs wavelet-based denoising, fuzzy-logic and neural-network classifiers, and multi-peak detection algorithms to separate atmospheric signal from interference. These can be very effective, but they are often complex, require tuning or training data, and may not transfer easily from one radar, frequency or site to another.

In summary, no single technique is satisfactory under all conditions. Most are tuned to a particular radar and climatology, and many were developed for 50 MHz or 400 MHz systems. The 205 MHz radar at CUSAT is relatively new, operates in a humid tropical environment with a strong monsoon cycle, and has its own noise characteristics and interference environment. Existing methods therefore need to be examined and reevaluated for this system, rather than adopted without question.

1.6 Problem Addressed in This Work

The problem addressed in this report is the reliable estimation of the Doppler spectral moments, and especially the location of the spectral peak, from the 205 MHz ST radar at CUSAT. Because the first moment gives the radial velocity, a failure to identify the correct peak translates directly into an erroneous wind. The objective is to obtain estimates that are accurate in low signal-to-noise conditions, resistant to contamination from clutter, interference and precipitation, and efficient enough for routine processing of the continuous data stream. To this end, two principal algorithms were investigated and compared:

1. Improved Moment Estimation (IME). This method refines the conventional moment calculation by objectively delimiting the portion of the spectrum that belongs to the atmospheric signal and by treating the noise level in a more careful way, so that the moments are computed only over the genuine signal region. It is computationally light and well suited to routine processing, and its behaviour on weak and contaminated spectra is of particular interest.
2. MCCF. This is a correlation-function-based approach that identifies the signal peak by exploiting the coherence and consistency of the signal across the data, instead of relying only on the amplitude of individual spectral bins. It offers a different principle of operation from IME and hence a useful basis of comparison.

The two methods were applied to the radar data and compared in terms of their accuracy, their robustness in the presence of noise and interference, and their computational cost. The aim of the comparison is not merely to name a “winner”, but to understand the circumstances under which each method succeeds, where it breaks down, and how the two may complement each other.

Chapter 2:
Estimation and Verification Methods

Various techniques for moment/peak estimations have been devised in the past decade. Some of them cover general cases while others are better suited for specific environments. In this chapter, we discuss a generic method and 2 established ones: MCCF and IME. We also discuss mixing IME and MCCF methods. In the 2nd part, we discuss some methods used to verify the accuracy of algorithms.

2.1 Estimation Methods

2.1.1 Mutual Convolution Cost Function (MCCF)

MCCF is a peak estimation algorithm proposed by Tang et al. [2] which uses Mutual convolution to improve SNR and a velocity-frequency-weighted cost function to estimate the most prominent peak in the spectrum. It also uses a space-time Doppler window (ST window) to limit the number of candidate peaks as well as to improve the continuity of the velocity profile. The space window is used to improve velocity continuity over height ranges, and the time window is used to improve velocity continuity in specific height over time.

The algorithm is illustrated in Figure 2.2. Here, the cylindrical entries are final data as well as intermediate data acquired. The intermediate data might be useful for other algorithms to work on in further processing steps.

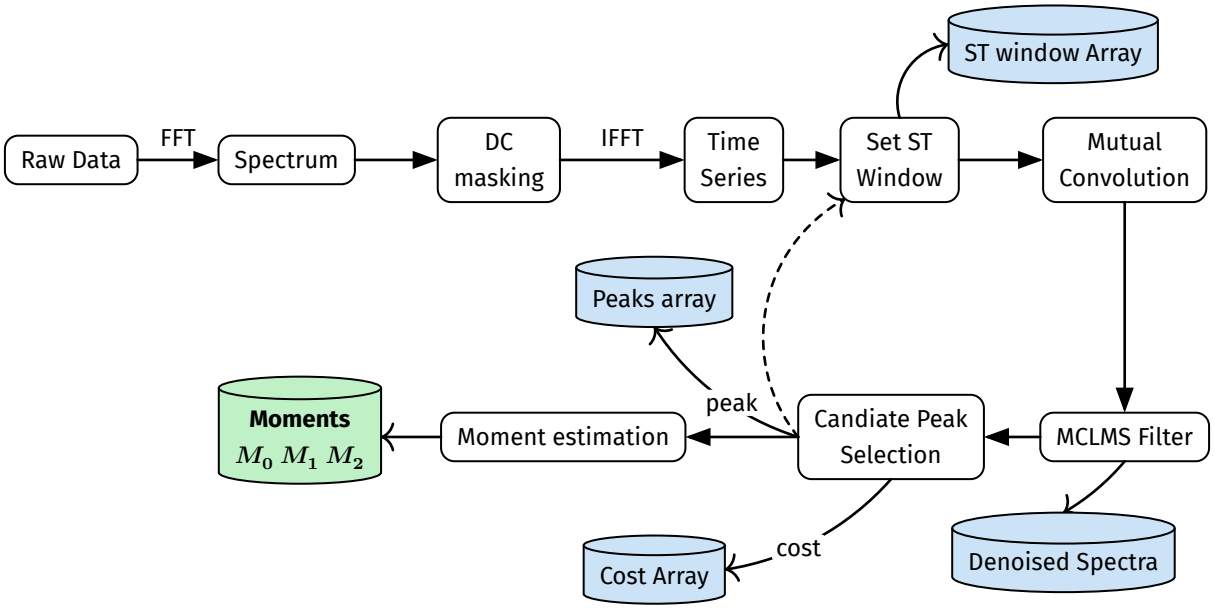


Figure 2.2: MCCF Algorithm Flow.

Individual steps are explained in upcoming subsections. The method applied to a doppler spectrum is shown in Figure 2.3

Preprocessing

Here the FFT of time series raw data (slow sampled) is computed. Then DC masking is applied in which value of 5 bins near DC bin is replaced with bacground noise with random phase. It is then converted back to time series data.

Setting the Space-Time Doppler Window

This step improves velocity profile continuity by limiting number of candidate peaks. By setting a doppler window in *space axis* peaks outside this window is rejected. This window accounts for maximum wind shear between height bins. Detting a doppler window in *time axis* applies the assumption that during a time interval wind velocity changes at some maximum rate. The Space-Time window is intersection of both windows.

Mutual Convolution

In this step the noise is heavily suppressed and SNR is improved. It works by slicing a spectrum in a height bin P into n slices of length $N/2$ (where N is length of original spectrum) then convolving adjacent slices ($X_i = V_i * V_{i+1}$) together to form $n-1$ results. From these results a optimal solution segment X_r is chosen which contain the highest peak among all results. Also the segment adjacent to X_r is chosen as suboptimal solution X'_r. Both of these are used in MCLMS filter.

MCLMS Adaptive Filtering

MCLMS (Modified Complex Least Mean Square) filter is a adaptive filter. It learns its weight during slicing of spectrums, and after modifying the weight vector final filtered spectra is obrained through the optimal and suboptimal spectras. MCLMS can suppress clutter clutter even more by training a model that tries to match optimal solution with suboptimal solution. In the end the spectrum will have some of the clutter suppressesd.

2.1.2 Candidate Peak Selection, Cost Function & Moment Estimation

Peaks in ST doppler window which has amplitude $\geq 40\%$ of maximum peak are chosen as candidate peaks. The algorithm assigns a "cost" to each of these peaks based on their velocity difference between each frequency bin $V_{r+1,j}$ given by:

$$V_{r+1,j} = 1 - \left| (M_{r+1,j} - M_r) \frac{d_v}{\alpha V_d} \right|$$

and magnitude of peak itself $M_{r+1,j}$. Here d_v represents the velocity resolution, V_d represents the maximum wind shear, α is the window coefficient. The cost of each peak is given by:

$$F_{r+1,j} = w_1 V_{r+1,j} + w_2 M_{r+1,j}$$

The peaks with higest cost value is taken as the spectral peak. From this moments can be estimated based on Hildreth-Seikh method.

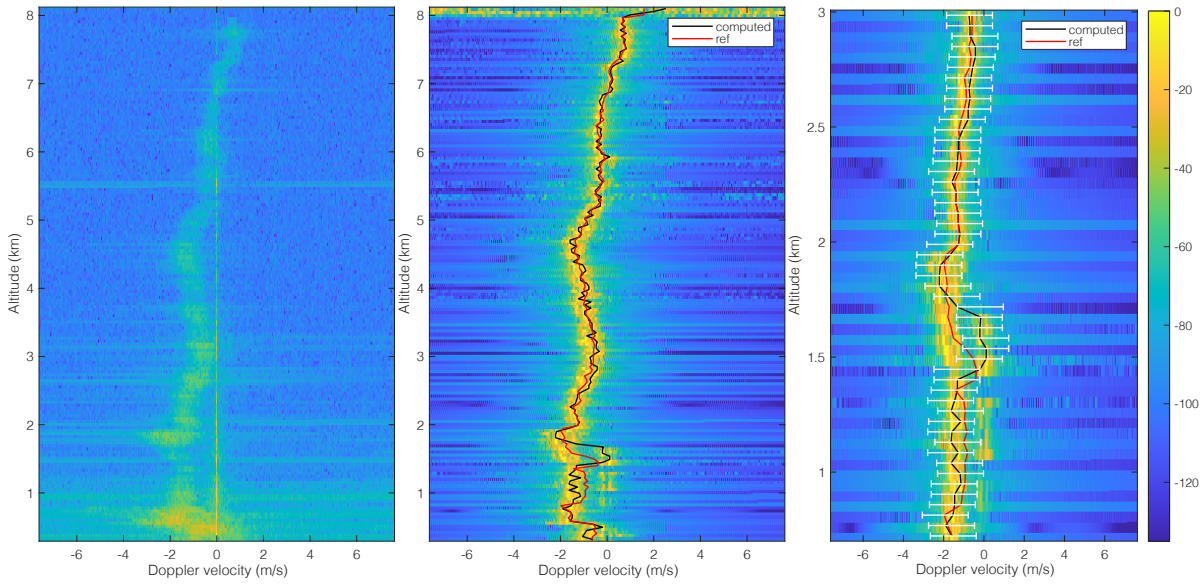


Figure 2.3: Power spectrum (a) Original, (b) after applying DC removal, mutual convoluton and MCLMS, (c) Computed moments, reference moment and ***Space window [white]*** *overlayed*

The peaks estimated from one height bin is used to set ST window in next height bin, hence the process is iterative and cannot be done in single matrix operation step as in computing spectrum.

2.1.3 Improved Moments Estimation (IME)

The Improved Moments Estimation (IME) algorithm proposed by Li **et al.**[3] is designed to extract reliable Doppler velocity profiles from ground-based K-band wind radar spectra, particularly when the atmospheric echoes have low SNR and are contaminated by clutter or multiple spectral peaks. The method combines the adaptive Doppler-window concept of the Adaptive Moments Estimation (AME) method with the cost-function-based profile tracing concept of the Multiparameter Cost Function (MPCF) method. Its main objective is to reduce the computational search space while maintaining reliable wind-profile detection under low-SNR conditions.

The IME procedure consists of three main stages:

1. Preprocessing of the Doppler power spectra
2. Adaptive Doppler-window selection and identification of prospective spectral peaks
3. Grouping, profile tracing, and cost allocation.

Data Preprocessing

Before profile extraction, the Doppler power spectra are processed to suppress unwanted spectral components. First, the zero-frequency region is affected by direct-current (DC) components and ground clutter. These components are suppressed by interpolation, in which the average values of three points around zero frequency are used to replace the contaminated values. Next, a five-point moving-average filter is applied to suppress transient interference while preserving the atmospheric echo characteristics. Finally, the noise power density S_n of each range bin is estimated using the Hildebrand–Sekhon method. The estimated noise level is subtracted from

the power spectrum, and negative values are set to zero. The resulting spectrum is therefore more suitable for identifying atmospheric spectral peaks. The preprocessing procedure described in the paper is summarized in Figure 2.4

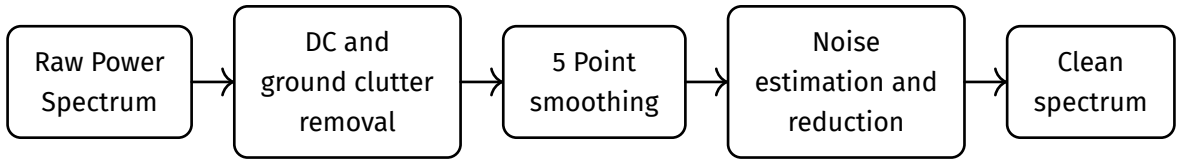


Figure 2.4: IME Method

These preprocessing operations help identify weak atmospheric echoes by reducing the effects of random noise, radio-frequency interferenc.,

Adaptive Doppler Window and Prospective Peak Selection

The IME algorithm starts at the lowest range bin and searches for candidate spectral peaks only within an adaptive Doppler window. For a selected spectral component, the algorithm determines the peak position and the left and right zero-crossing points, from which the radial velocity, spectral width, and SNR are calculated. A peak is retained as a prospective atmospheric component when its SNR is greater than 0 dB. Up to three prospective peaks are retained for each range bin. The procedure stops earlier when no spectral component with power greater than the estimated noise level remains in the current Doppler window.

For the first range bin, no previous information is available. Therefore, the most prominent spectral component is initially selected and its radial velocity is used as the mean Doppler velocity. The Doppler window for the next range bin is then centered around the candidate whose radial velocity is closest to the previous mean Doppler velocity. This process is repeated sequentially with increasing range.

The adaptive-window procedure is represented in Figure 2.5

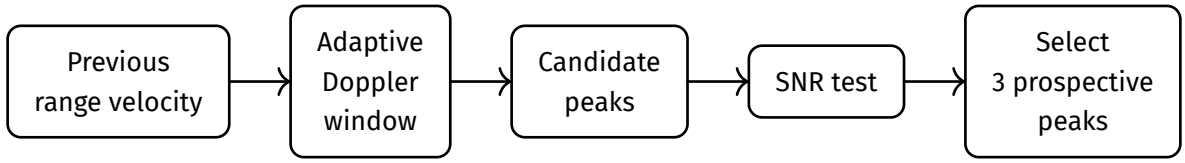


Figure 2.5: IME Adaptive window procedure

The wind Doppler signal is generally not expected to change by more than 20% of the total coherent-filter bandwidth, and therefore the Doppler-window width is restricted to less than 20% of the complete Doppler-axis width.

Grouping and Cost-Function-Based Profile Tracing

In this step the algorithm constructs possible traces between adjacent range bins. A trace represents a possible Doppler velocity profile through height. Constructing all possible traces over the complete range would result in a large computational load; therefore, the spectra are divided into small groups containing at least three consecutive range bins. Candidate peaks within each group are then connected from one range bin to the next. For the first group, the initial value is the Doppler velocity of the most prominent peak, while subsequent groups use

the velocity at the highest range bin of the preceding group as their initial value. Figure 2.6 shows all links in a group and M_j^r indicates the j th peak of the r th range bin.

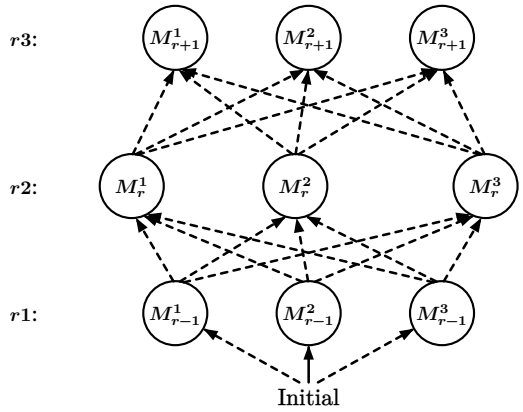


Figure 2.6: Grouping and profile tracing in IME. Each circle represents a candidate spectral peak

The cost assigned to a connection between two candidate peaks is based on two characteristics: the continuity of wind shear and the relative SNR of the candidate. The maximum allowable radial-velocity shear is determined by;

$$|\Delta v_{\mathrm{m}}| \leq 0.05 \sin(\theta) \Delta R \tag{2.1}$$

where Δv_{m} is the maximum radial-velocity shear, θ is the radar beam tilt angle, and ΔR is the range resolution.

To represent the continuity of the Doppler profile, a trapezoidal membership function is used:

$$T(x) = \begin{cases} 1 & |x| \leq \Delta v_{\mathrm{m}} \\ (2\Delta v_{\mathrm{m}} - |x|)/\Delta v_{\mathrm{m}} & \Delta v_{\mathrm{m}} < |x| < 2\Delta v_{\mathrm{m}} \\ 0 & |x| \geq 2\Delta v_{\mathrm{m}} \end{cases} \tag{2.2}$$

Here, x represents the wind-shear difference between two connected candidate peaks. A high membership value indicates that the change in velocity between adjacent range bins is consistent with the expected continuity of the atmospheric wind profile.

The relative SNR of a candidate peak is normalized with respect to the largest candidate SNR in the same range bin:

$$P_{r,j} = \frac{\alpha_{r,j}}{\max(\alpha_r)} \tag{2.3}$$

where $\alpha_{r,j}$ denotes the SNR of the j-th candidate peak in the r-th range bin. The complete link cost is then formed by combining the continuity term and the normalized SNR term:

$$C = w_1 T + w_2 P, \quad w_1 + w_2 = 1. \tag{2.4}$$

The weights determine the relative importance of wind-profile continuity and spectral strength. Based on experiments the selected values are,

$$w_1 = 0.4, \quad w_2 = 0.6.$$

Thus, the IME algorithm gives greater weight to the normalized SNR while still enforcing continuity between adjacent range bins. The total cost of each possible trace is calculated, and the trace with the largest total cost is selected as the final Doppler velocity profile for the corresponding group. The velocity at the highest range bin in the selected group then becomes the initial value for the next group.

Thus, Figure 2.7 illustrates how IME narrows the peak-search region and identifies a more consistent atmospheric Doppler profile.

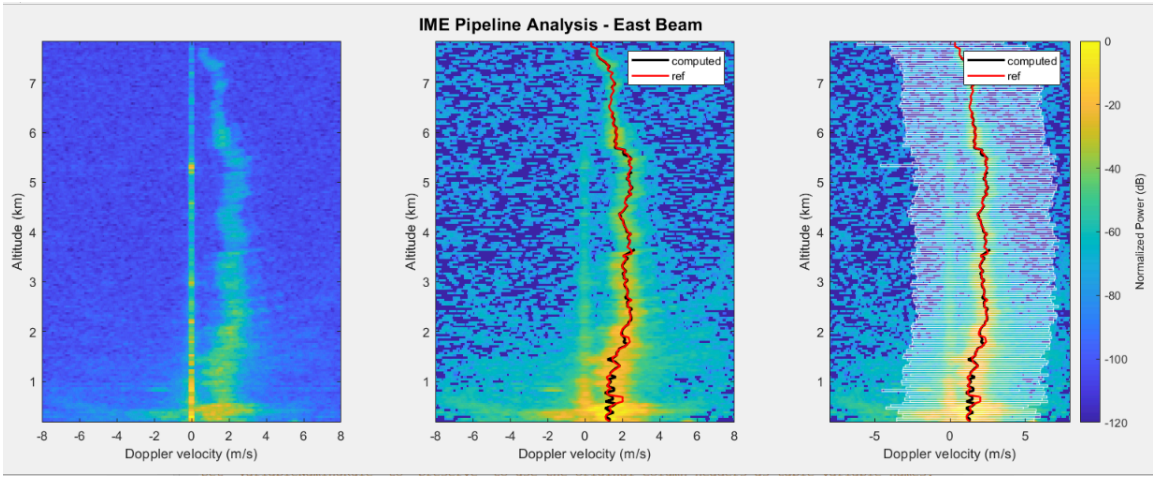


Figure 2.7: Doppler beam spectra a) Original b) after IME processing, c) Adaptive Doppler-window selection.

2.1.4 Hybrid Methods

The 2 algorithms discussed have several individual working components – computing spectrum, denoising, estimating peaks/moments, etc. The mutual convolution process in MCCF appears to suppress noise very effectively. In this light, we tried a method that combines both of these methods.

To make such mixing easier, our main focus was to first create a software architecture in MATLAB that encapsulates each of these components and provides a common interface. Then the individual components can be cascaded easily. Figure 2.8 Shows the method in which Mutual Convolution (MC) from MCCF is first used to determine the spectrum, and then later processing is done by IME itself.

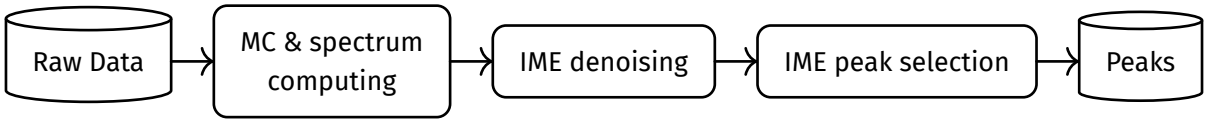


Figure 2.8: Flow of Hybrid-1 method

Figure 2.9 shows another method in which we eliminated denoising part of IME (since MC already performs some amount of denoising, we suspect that further denoising eliminates candidate peaks in low SNR regions)

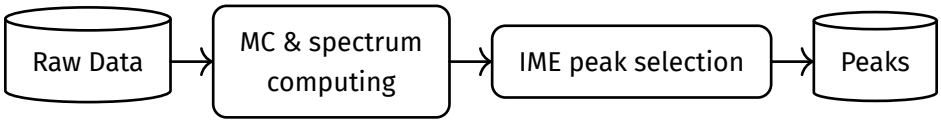


Figure 2.9: Flow of Hybrid-2 method

In practice, the parts of MCCF work iteratively in each height bin, taking feedback from previous bin, setting ST window appropriately, then selecting the best solution from MC that contains a peak falling in the ST window. For this mixing purpose, we took the MC part alone and let it run without the feedback.

2.2 Verification methods

To verify the algorithms we need a reliable ground truth to compare against the output of the algorithm. Below are few methods generally used in literature.

2.2.1 Radiosonde

A radiosonde is a battery-powered telemetry instrument carried into the atmosphere usually by a weather balloon that measures various atmospheric parameters and transmits them by radio to a ground receiver. Modern radiosondes measure or calculate the following variables: altitude, pressure, temperature, relative humidity, wind (both speed and direction), and some are also capable of cosmic ray readings at high altitude.



Figure 2.10: Radiosonde attached to weather balloon

The radiosonde reaches an altitude of 20 km to 30 km; thereafter, due to the low density of the atmosphere, the balloon explodes, and it falls down. The altitude reached in various radiosonde experiments is plotted in Figure 2.11 data = data[::30]

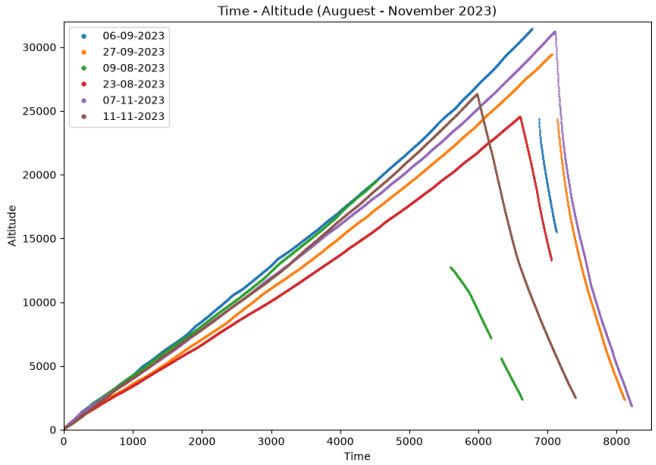


Figure 2.11: Altitude reached by various radiosonde.

In order for the data from the radiosonde to be useful for evaluating the algorithm, the balloon must hover over the same geographic area throughout the desired height range. For perspective, the ST radar has a beamwidth of 3°, Considering 30km target height, ideally, the ballon should be within $\tan(3°/2) \times 30 \text{ km} \approx 1 \text{ km}$ radius within the launch site. But this can be much higher, considering that the atmosphere can be slowly varying in certain seasons. However, the radiosondes often drift too far from their initial destinations. This is illustrated in Figure 2.13 and Figure 2.12. Data collected in the summer season is used for testing the algorithm. However, in those seasons, wind velocities are not turbulent enough to rigorously test the algorithm.

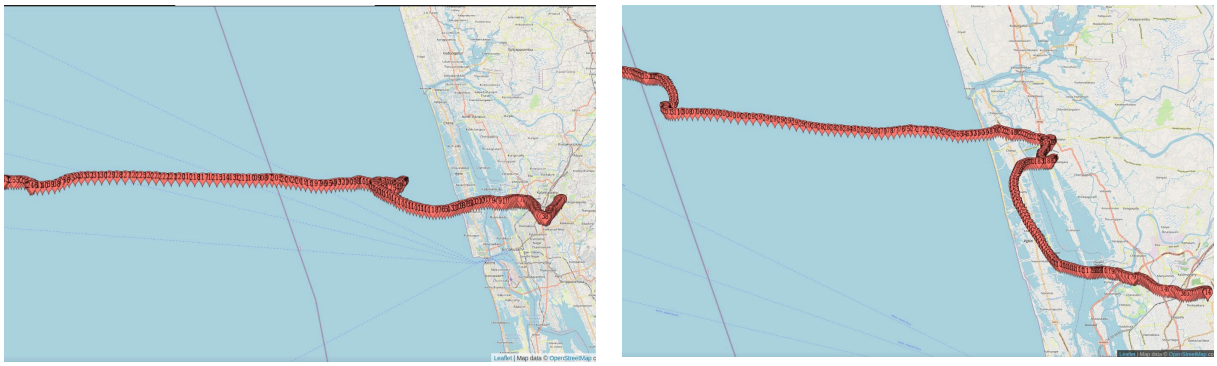


Figure 2.12: Trajectory of Radiosonde in the month of November 2023

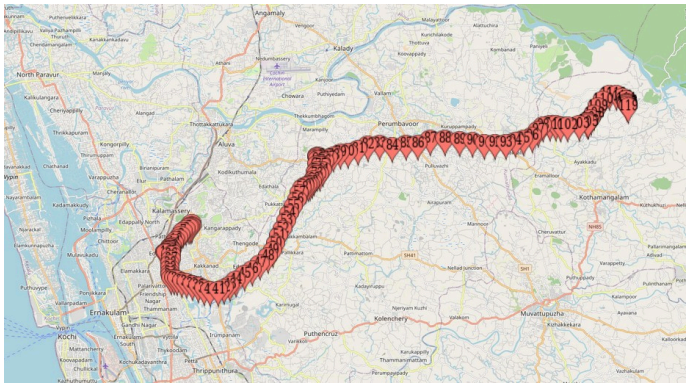


Figure 2.13: Trajectory of Radiosonde in the month of February 2021

2.2.2 LIDAR

Light Detection and Ranging (LIDAR) is a method for determining ranges by targeting an object or a surface with a laser and measuring the time for the reflected light to return to the optical receiver. Lidar may operate in a fixed direction (e.g., vertical), or it may scan directions, in a special combination of 3D scanning and laser scanning. In [2] and [3], the researchers used a ground-based lidar for measuring the wind velocity of the lower atmosphere (up to 1 km). In [2], it is mentioned that the lidar can achieve 99% accuracy in clear atmospheric conditions. Hence, lidar serves as an excellent ground reference for evaluating the algorithm's performance.



Figure 2.14: A Lidar mounted on the rooftop of a building

2.2.3 Synthetic data

Synthetic radar data are artificially generated radar observation data made from a predefined array of wind velocities. This is done by starting with a UVW array and reversing the process of a typical radar signal processing pipeline, starting from UVW estimation to writing a raw file onto disk. The primary advantage of this method is that we have a ground truth to compare the algorithm's output as well as control to create data that can test the algorithms in various ways, which otherwise would require waiting to get suitable outputs from radar and manually going through them to pick a good one. This method is discussed depth in *Chapter 3: Synthetic data creation*

Chapter 3:
Synthetic data creation

While the algorithms are being developed, it is important to measure their performance with some metric. The metrics can include latency, computational efficiency, accuracy, etc. For ST radar, which completes a full observation across 5 beams every 80 seconds, the first 2 seem irrelevant. To measure accuracy, either as MSE or correlation

3.1 Result

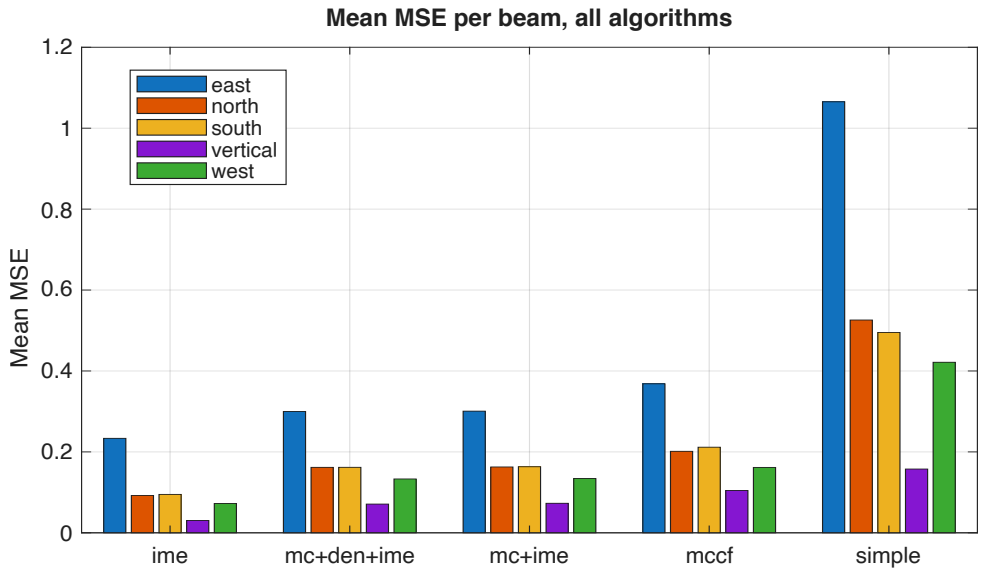


Figure 3.15: Comparison of MSE of various algorithms

References

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